Mobile-Wallet Fraud Detection

A cost-aware screening model for PaySim transactions

Machine Learning Final Project · NextGen Institute

Tuned random forest | Streamlit demo | Simulated data

Fraud detection is a decision problem

A score only matters when it helps the provider choose what to do next.

TRANSACTION

Transfer or cash-out

Available at decision time:
type, amount, balances, hour



FRAUD SCORE

0 to 1

Pipeline turns transaction fields into a probability.



ACTION

ALLOW
REVIEW
BLOCK

The trade-off: missed fraud costs the stolen amount; false alarms disrupt honest customers.

The data is large and highly imbalanced

PaySim contains millions of simulated mobile-money transactions.

6.36M
transactions

8,213
fraud cases

0.13%
of all transactions

Class balance



99.87% honest **0.13% fraud**

A model that predicts “honest” for every row still reaches 99.87% accuracy.

Only decision-time signals belong in the model

Fraud appeared in TRANSFER and CASH_OUT; post-transaction fields were excluded.

INPUTS

Transaction fields

type · amount · sender balance
receiver balance · hour of day

ENGINEERED SIGNALS

Sender drained · amount / balance
Receiver was empty

LEAKAGE CONTROLS

Excluded from training

Balance-after fields: known only after processing

Simulator flag: sparse and rule-derived

Customer IDs: near-unique identifiers

The app only asks for values available before the decision.

A single pipeline keeps preprocessing inside validation

The test set stayed untouched until the final evaluation.

STRATIFIED SPLIT

80%

training data

20%

held-out test

TRAINING PIPELINE

Feature creation

Then log-scale numeric fields, standardize, and one-hot encode type.

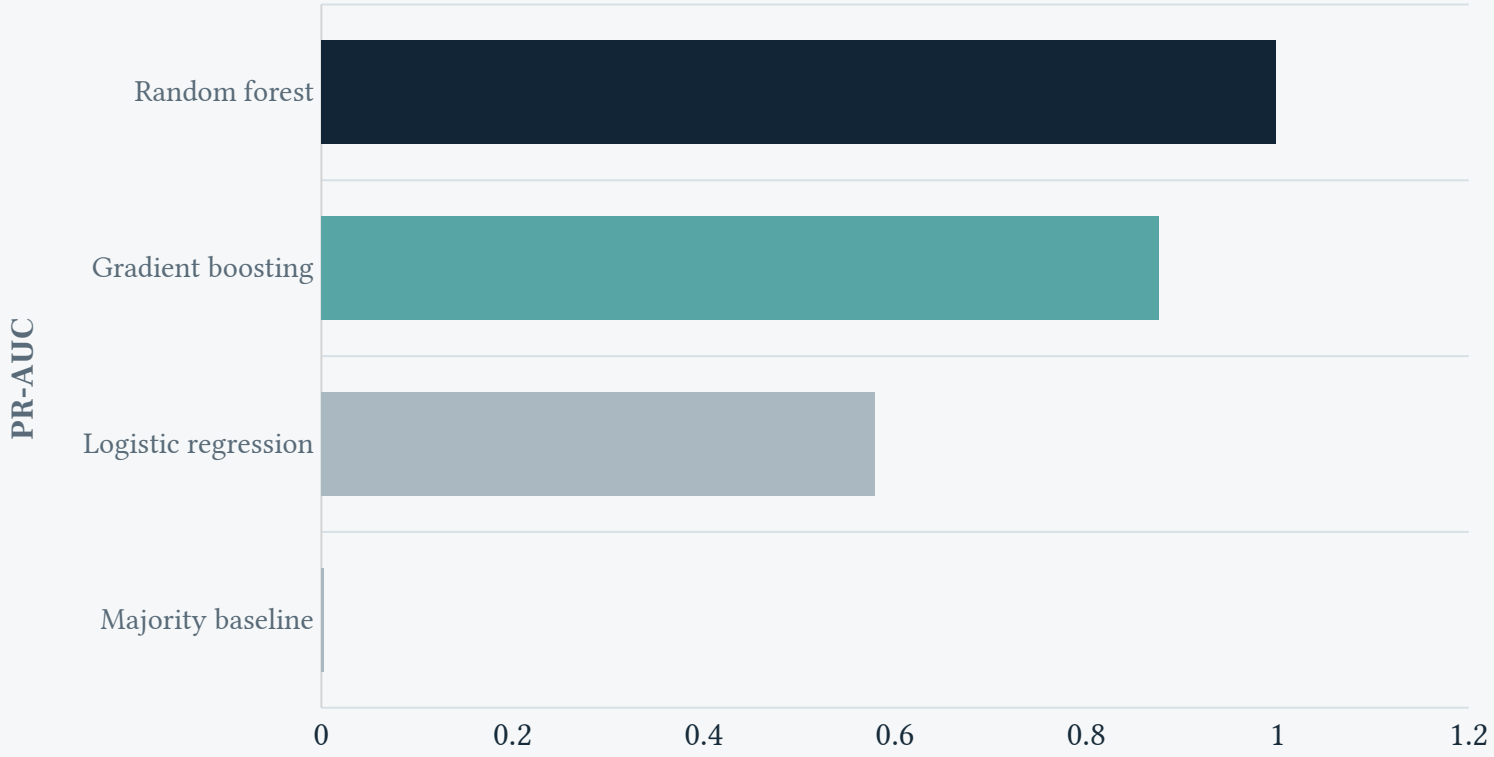
Model selection

Stratified 3-fold cross-validation and randomized search use training data only.

The final score comes from the separate test set.

Engineered features lift PR-AUC across the model comparison

Three-fold cross-validation on the development sample; higher is better.



RANDOM FOREST

0.998

The same forest scored 0.817 with raw columns only.

Feature engineering added more than model choice alone.

Tuning selected a stable random forest

Randomized search used three-fold cross-validation on training data.

Model	CV PR-AUC	Fold std	Decision
Tuned random forest	0.9985	0.0016	Selected
Tuned gradient boosting	0.9913	0.0060	—
Default random forest	0.9978	—	Reference

SELECTED SETTINGS

200 trees · max depth 20 · minimum leaf size 10 · max_features = sqrt

Held-out test performance is strong

One final evaluation on 554,082 transactions, including 1,643 fraud cases.

PR-AUC

0.998

ROC-AUC

0.9991

PRECISION AT 0.50

0.9994

RECALL AT 0.50

0.9963

CONFUSION MATRIX · THRESHOLD 0.50

	Predicted honest	Predicted fraud
Actually honest	552,438	1
Actually fraud	6	1,637

1,637 / 1,643

frauds caught at 0.50

The remaining errors break the model’s learned pattern

Missed frauds often did not drain an account or target an empty destination.

MISSED FRAUD

6

3 TRANSFER · 3 CASH_OUT

At least 4 of 6 did not empty the sender account; at least 4 went to a destination that was not empty.

ONE FALSE ALARM

1.08M

Honest transfer from a zero recorded sender balance to an empty destination.

The model learned a simple simulator pattern. Fraud that differs from it can be confidently missed.

Lowering the threshold cannot recover all misses without raising more honest alerts.

Review threshold and false-alarm cost

Cost curve uses out-of-fold training predictions, not the test set.



ASSUMED FALSE-ALARM COST

100 units

BEST TRAINING THRESHOLD

0.02

3,586 false alarms
6.81M missed fraud amount
7.17M total cost

At false-alarm costs of 10,000 or more, the best threshold shifts to 0.60.

The Streamlit app turns scores into a review workflow

A local demo was tested with single transactions, CSV input, invalid input, and threshold controls.

01

Enter

Five decision-time fields

02

Score

Saved preprocessing + forest

03

Route

Allow / review / block

04

Inspect

Threshold costs and CSV results

LOCAL STATUS

App flows passed; public deployment is pending.

A strong simulator result is a starting point

WHAT THE MODEL SHOWS

Decision-time balance features make PaySim fraud highly separable.

The review policy makes the business cost assumption visible and adjustable.

WHAT MUST HAPPEN NEXT

Deploy the app
Validate costs with a real operations team
Test on representative, labeled wallet data
Monitor drift and retrain

PaySim is simulated; these metrics are not a real-world guarantee.

Dataset: PaySim Mobile Money Simulation · kaggle.com/datasets/ealaxi/paysim1